```
# Install Required Libraries
!pip install pandas numpy nltk scikit-learn

# Import Libraries
import pandas as pd
import numpy as np
import re
import string
import nltk

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

# Download NLTK Resources
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

try:
    nltk.download('punkt_tab')
except:
    pass

# Create Dataset
data = {
    "Text": [
        "I love this phone",
        "This mobile is excellent",
        "Amazing camera quality",
        "Very good battery",
        "I like this product",
        "I hate this phone",
        "Worst mobile ever",
        "Very bad battery",
        "Poor camera quality",
        "I dislike this product"
    ],

    "Sentiment": [
        "Positive", "Positive", "Positive", "Positive", "Positive",
        "Negative", "Negative", "Negative", "Negative", "Negative"
    ]
}

# Convert Dataset into DataFrame
df = pd.DataFrame(data)

print("========== ORIGINAL DATASET ==========\n")
print(df)

# ----------------------------
# TEXT PREPROCESSING
# ----------------------------

# Convert to Lowercase
df["Text"] = df["Text"].str.lower()

# Remove Punctuation
df["Text"] = df["Text"].str.replace(
    '[{}]'.format(string.punctuation),
    '',
    regex=True
)

# Remove Numbers
df["Text"] = df["Text"].str.replace(
    r'\d+',
    '',
    regex=True
)

# Tokenization
df["Text"] = df["Text"].apply(word_tokenize)

# Remove Stopwords
stop_words = set(stopwords.words("english"))
```

```
df["Text"] = df["Text"].apply(
    lambda words:
    [word for word in words if word not in stop_words]
)

# Stemming
stemmer = PorterStemmer()

df["Text"] = df["Text"].apply(
    lambda words:
    [stemmer.stem(word) for word in words]
)

# Lemmatization
lemmatizer = WordNetLemmatizer()

df["Text"] = df["Text"].apply(
    lambda words:
    [lemmatizer.lemmatize(word) for word in words]
)

# Join Words
df["Text"] = df["Text"].apply(
    lambda words: " ".join(words)
)

print("\n========== PROCESSED DATASET ==========\n")
print(df)

# ----------------------------
# FEATURE EXTRACTION
# ----------------------------

vectorizer = CountVectorizer()

X = vectorizer.fit_transform(df["Text"])

y = df["Sentiment"]

# ----------------------------
# MODEL TRAINING
# ----------------------------

model = MultinomialNB()

model.fit(X, y)

print("\n========== MODEL TRAINED SUCCESSFULLY ==========")

# ----------------------------
# TEST SENTENCE
# ----------------------------

sentence = "This phone has good camera"

# Preprocessing the Input Sentence
sentence = sentence.lower()
sentence = re.sub(r'[^\w\s]', '', sentence)
sentence = re.sub(r'\d+', '', sentence)

words = word_tokenize(sentence)

words = [
    word for word in words
    if word not in stop_words
]

words = [
    stemmer.stem(word)
    for word in words
]

words = [
    lemmatizer.lemmatize(word)
    for word in words
]

sentence = " ".join(words)

# Convert Text into Vector Form
test = vectorizer.transform([sentence])
```

```
# Predict Sentiment
prediction = model.predict(test)

# Display Output
print("\n========== TEST RESULT ==========\n")
print("Test Sentence :", sentence)
print("Predicted Sentiment :", prediction[0])
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.3)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.4.2)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
========== ORIGINAL DATASET ==========

                       Text Sentiment
0         I love this phone  Positive
1  This mobile is excellent  Positive
2    Amazing camera quality  Positive
3         Very good battery  Positive
4       I like this product  Positive
5         I hate this phone  Negative
6         Worst mobile ever  Negative
7          Very bad battery  Negative
8       Poor camera quality  Negative
9    I dislike this product  Negative

========== PROCESSED DATASET ==========

                  Text Sentiment
0           love phone  Positive
1          mobil excel  Positive
2  amaz camera qualiti  Positive
3         good batteri  Positive
4         like product  Positive
5           hate phone  Negative
6     worst mobil ever  Negative
7          bad batteri  Negative
8  poor camera qualiti  Negative
9       dislik product  Negative

========== MODEL TRAINED SUCCESSFULLY ==========

========== TEST RESULT ==========

Test Sentence : phone good camera
Predicted Sentiment : Positive
```